

```

117.     colors[0] = headTailColor # Set head color.
118.     colors[-1] = headTailColor # Set tail color.
119.
120.     # Set up the rest of fish data structure:
121.     fish = {'right':      fishType['right'],
122.            'left':       fishType['left'],
123.            'colors':     colors,
124.            'hSpeed':     random.randint(1, 6),
125.            'vSpeed':     random.randint(5, 15),
126.            'timeToHDirChange': random.randint(10, 60),
127.            'timeToVDirChange': random.randint(2, 20),
128.            'goingRight':  random.choice([True, False]),
129.            'goingDown':   random.choice([True, False])}
130.
131.     # 'x' is always the leftmost side of the fish body:
132.     fish['x'] = random.randint(0, WIDTH - 1 - LONGEST_FISH_LENGTH)
133.     fish['y'] = random.randint(0, HEIGHT - 2)
134.     return fish
135.
136.
137. def simulateAquarium():
138.     """Simulate the movements in the aquarium for one step."""
139.     global FISHES, BUBBLERS, BUBBLES, KELP, STEP
140.
141.     # Simulate the fish for one step:
142.     for fish in FISHES:
143.         # Move the fish horizontally:
144.         if STEP % fish['hSpeed'] == 0:
145.             if fish['goingRight']:
146.                 if fish['x'] != RIGHT_EDGE:
147.                     fish['x'] += 1 # Move the fish right.
148.             else:
149.                 fish['goingRight'] = False # Turn the fish around.
150.                 fish['colors'].reverse() # Turn the colors around.
151.         else:
152.             if fish['x'] != LEFT_EDGE:
153.                 fish['x'] -= 1 # Move the fish left.
154.             else:
155.                 fish['goingRight'] = True # Turn the fish around.
156.                 fish['colors'].reverse() # Turn the colors around.
157.
158.         # Fish can randomly change their horizontal direction:
159.         fish['timeToHDirChange'] -= 1
160.         if fish['timeToHDirChange'] == 0:
161.             fish['timeToHDirChange'] = random.randint(10, 60)
162.             # Turn the fish around:
163.             fish['goingRight'] = not fish['goingRight']
164.
165.         # Move the fish vertically:
166.         if STEP % fish['vSpeed'] == 0:
167.             if fish['goingDown']:
168.                 if fish['y'] != BOTTOM_EDGE:
169.                     fish['y'] += 1 # Move the fish down.
170.             else:
171.                 fish['goingDown'] = False # Turn the fish around.

```